Appendix: Worked brush-graph thought experiments

These are **candidate** signal paths for a future brush graph, not implemented Painter nodes, presets, or promises of visual equivalence. The brush graph produces marks or material operations **inside a canvas**; a separate canvas-composition graph would combine canvas outputs. A socket below carries a value **and** a domain/clock: `StrokeSamples@stroke`, `DabEvent@dab`, `MaskField@dab/pixel`, `ImageRead@revision`, and `MaterialState@tick` cannot be connected as if they were interchangeable colors. Values below are **illustrative experiment settings, not defaults**. See "Proposed typed vocabulary" in the brush-engines section and "Execution phases, contracts, and the stroke boundary" in graph-systems. References in those parent sections are numbered locally, not across the report.

For all paths, a possible **future** durable command is ordered input samples plus brush-graph/version and asset identities, seed, chosen color policy, and any required time/source/solver parameters; emitted marks, rendered tiles, and rebuildable checkpoints are derived caches. This is a proposed replay contract, **not** the present file format. Today Painter saves only immutable ordered document-space points and weights; it deterministically derives pencil splats and caches raster tiles. Editing or undoing an old stroke means regenerating affected outputs in order; a cache image is never authoritative input merely because it happens to be on screen. The important difference between examples is **how far** dependency invalidation must travel.

A. Current-style pencil: local, additive

`StrokeSamples@stroke (position: document length, weight: unit fraction) → SeededSpacing@stroke (e.g. 1.5 document units) → DabEvent@dab (position, size, stable index) → GaussianSpecks@dab (seeded jitter, e.g. 6 specks/dab at full weight) → MaskField@dab/pixel (coverage) → Deposit@dab (fixed RGB, coverage) → MarkStream@stroke → raster tiles@resolution`.

This is a **hypothetical** 1.5-unit resampler, not Painter's pencil algorithm: today `SplatStream.extend` emits one sample at the first point and `max(1, ceil(segment length / 0.5))` evenly spaced samples per subsequent segment, ending at each point. Each sample emits `max(1, round(6 × interpolated weight))` Gaussian splats scattered in a disc of radius `brushDiameter × weight / 2`; weight changes both radius and speck count, not just size. Mouse weight is `1.0`; stylus weight follows the sampled input. Present speck randomness hashes stroke seed, sample index, and speck index, retaining a stable prefix on append. The proposed graph likewise needs stable dab/speck indices to preserve that property, but its 1.5-unit spacing and output are not claims of current visual equivalence. Its result is a bounded, ordered additive mark stream. Today only sampled points/weights persist; Painter derives a stroke seed from document order when loading, so deleting/reordering strokes would change their grain. A future editable-stroke graph would persist stable stroke identity/seed plus graph identity/settings. Derived specks and tile pixels remain cached. Editing/undoing an older stroke invalidates intersecting cached tiles and replays their ordered surviving marks; later **pure** strokes do not acquire new colors. Painter's present incremental additive tile update is useful for appends, not proof that an edited old stroke can be erased from the old tile. See brush-engines "Candidate mappings: paths, controls, and the honest boundary" (graphite/pencil) and `docs/architecture.md` ("Canvas Architecture").

B. Two textured dual-brush routes, not one operator

Intersected tips, Photoshop-inspired: `StrokeSamples@stroke → Spacing@stroke (e.g. 4 document units) → DabEvent@dab`; branch A `TipA@dab (soft ellipse) × PaperTexture@document/pixel (e.g. fixed paper scale 12 document units) → MaskA@dab/pixel`; branch B `SecondaryEmitter@dab (e.g. 2 seeded offsets per A dab) →`

`TipB@secondary-dab/pixel (irregular tip) → MaskB@dab/pixel`; `Intersect(MaskA, MaskB)@dab/pixel → Deposit(ColorRGB, coverage)@dab → MarkStream@stroke`. The secondary emitter must explicitly gather its masks back into A's dab region before the pixelwise intersection; `MaskB` is **not** an independent stroke composited on top. Photoshop documents dual-brush intersection alongside texture and its own secondary spacing/scatter; this sketch does not reproduce its equations (Adobe dual brush/texture).

Two complete brush outputs, Procreate-style composition: `StrokeSamples@stroke → BrushA@stroke (own emitter, grain, color, MarkStreamA) + BrushB@stroke (own emitter, grain, color, MarkStreamB) → Rasterize(A, B)@tile/resolution → Blend(A image, B image; e.g. multiply)@tile/resolution → ImageResult@tile`. This is an **image-combination** boundary, not a tip-mask intersection; the blend order, alpha/color space, and whether the two outputs share a source snapshot must be declared. Procreate documents combining two brushes with blend modes, not Photoshop's tip-intersection algorithm (Procreate Dual Brush). The two paths above are alternatives, not claims that either app uses these exact graph internals.

For both experiments, persist input, separate graph/asset identities, seeds, texture anchoring, and blend/coverage policy; cache masks, streams, and resulting tiles. Neither reads earlier canvas pixels. Editing/undoing an earlier stroke invalidates tiles it covered and recomposites surviving ordered marks; it need not reevaluate later brush **inputs** unless an explicitly declared canvas read is added. A document-fixed texture is sampled at document coordinates, not restarted at every tile edge. See brush-engines "Candidate mappings" (textured dry brush/dual brush).

C. Clone or smudge: read scope and previous-dab state

Clone variant: `StrokeSamples@stroke + SourceAnchor@stroke (document position) + SourceScope@stroke (canvas ID or composite projection ID) + ReadPolicy@stroke (ordered pre-stroke or retained snapshot) → DabEvent@dab → SampleImage(ImageRead@declared-source, source = anchor + dab displacement)@dab/pixel → ColorField@dab/pixel × TipMask@dab/pixel → Deposit@dab → MarkStream@stroke`. Scope chooses **which** canvas or composition projection is read; it does not freeze its contents. With an ordered pre-stroke read, sample the selected source as reconstructed immediately before this stroke in document order (the clone cannot see its own dabs); an earlier source edit changes the read and forces dependent replay. A deliberately frozen read instead requires retained pixels or a fully reconstructable pinned snapshot, **not** just a revision label; neither mode is a current Painter clone API. A stale display tile must not silently substitute for the declared source. If the source is another canvas in the composition graph, identify and version the projection and guard against a dependency cycle.

Smudge variant: `DabEvent[n]@dab + ImageRead@pre-dab[n] (destination neighborhood) + Reservoir[n]@dab (loaded RGB, amount) → SampleImage@dab/pixel → PickupMix@dab (e.g. pickup fraction 0.2) → Deposit@dab + Reservoir[n+1]@next-dab`; commit the dab, then form `ImageRead@pre-dab[n+1]`. This is an explicitly ordered delay, **not** an instantaneous feedback loop. Reset/refill reservoir at stroke start, or persist a declared cross-stroke reservoir dependency. RGB smearing is an appearance model, not mass-conserving pigment transport. Krita distinguishes smearing from dulling and warns that sampling an all-layers overlay can make later lower-layer edits surprising (Krita Color Smudge); its clone source can be a layer or projection (Krita Clone).

Persist source anchor, source scope/identity and read policy; for ordered reads, retain the source command-order boundary and replay dependencies, and for frozen reads, retain the snapshot contents or every identity/version/input needed to reconstruct them. Persist graph/version, samples, seed, and the smudge reservoir initial/reset rule; cache sampled patches, intermediate reservoir

values, marks, and tiles. Undo/edit an earlier **source** stroke must invalidate an ordered-read clone and all dependent smudge dabs (and later dependent strokes), not only its geometric footprint. A deliberately retained frozen source keeps clone output stable across later source edits at a storage/reconstruction cost; merely recording a revision ID without retaining or reconstructing its contents is insufficient. See brush-paint "The tiled, replayable document is the hard interface."

D. Water/pigment: a clocked neighborhood, not just wet stamps

`StrokeSamples@stroke → DabEvent@dab → StampMask@dab/pixel → Deposit(WaterAmount, PigmentMass[palette])@cell/tick`; then on each declared `Duration@tick` (e.g. 1/60 s), `ReadMaterial@state[t] (water, mobile pigment, fixed pigment) + Halo@neighbor/tick → FluxAcrossEdges@tick → Absorb/Pin@tick → Dry(evaporation, fixation)@tick → state[t+1] → Shade@cell/pixel (stated optical model) → ImageResult@tile`. The brush graph supplies authored deposits and controls; a **canvas-owned** simulation clock owns the continuing state after pen-up. `Shade` converts material to display color; an RGB image cannot stand in for `PigmentMass[]`. Example water load `0.3` and dry rate `0.02 /s` are test inputs with units to be defined, not presets or measured physics. See brush-paint "Separate color optics from material motion" and "Feedback and typed candidate operations" (watercolor, MoXi, IMPaSTo references numbered there).

At each tick, read **one frozen prior state**, exchange same-tick neighbor halos, and compute one owner for each cross-tile flux edge. On an interior edge debit and credit the same pigment/water amount; report evaporation as water sink and fixation as mobile-to-fixed transfer. Dirty regions include brush radius, stencil/transport reach, and wet growth beyond originally touched tiles. State the canvas-edge boundary condition and solver cell size independent of display-tile resolution. Persist authored deposits, initial material policy, graph/material/solver versions, cell size, time-step sequence, and any promised external pigment assets; either rebuild material fields from commands or explicitly choose durable state as part of a **different** document model. Tile images and optional reproducible checkpoints are caches under the replay route. Editing/undoing an old deposit or drying tick invalidates later ticks and neighboring regions reached by transport; replay from a compatible checkpoint, never simply paint over the changed pixels. A fast coarse appearance preview versus a fine export may be visibly different unless the simulation resolution is fixed by the saved contract. See brush-paint "Simulation accounting versus a convincing picture" and "The tiled, replayable document is the hard interface."

E. Optional stationary airbrush: time is an input

`StrokeSamples@stroke + HoldTicks@time (e.g. 30 ticks/s while down) → RateEmitter@tick (e.g. 2 dabs/tick, seeded) → SoftMask@dab/pixel → LowFlowDeposit@dab (e.g. 0.01 coverage) → MarkStream@stroke`. Distance-only spacing would emit nothing while the pen is still; distinguish per-second flow from a stroke opacity cap. Persist timestamp/tick history or an explicitly sampled duration along with seeds and graph settings; neither time nor held ticks can be recovered from Painter's historical point/weight records. Cache emitted marks/tiles. An old stroke edit/undo requires recompositing affected tiles; later inputs remain independent unless a canvas read is added. See brush-engines "Candidate mappings" (soft airbrush; cited Krita/MyPaint sources).

Quick falsification experiment

- **Sensor absence:** Replay identical mouse and stylus traces with pressure/tilt/time missing in turn; a graph requiring an absent sensor should expose an unavailable port or declared fallback, never invent measurements. Historical strokes cannot be retrofitted with elapsed time.
- **Seams and scope:** Cross document-space tile edges with textured tips, clone source offsets, and a wet front; compare a one-region reference with tiled output for color continuity and water/pigment edge balance. Edit a source stroke outside the clone destination's footprint: an ordered-

read clone must invalidate dependent marks, while a clone using a retained frozen snapshot must retain its original source pixels.

- **Replay:** Save inputs/versions/assets/seeds/ticks, alter an early stroke, undo/redo, reopen, and compare regenerated marks and material fields to a clean ordered replay (pixel differences and conserved-quantity error, with specified tolerance). Run with preview thumbnails on/off to detect viewer-dependent evaluation.
- **Cost and fidelity:** On the same sampled paths, measure input-to-visible-mark latency (p50/p95), dabs/s, per-dab graph time, tile redraw/dirty area, simulation tick time, memory and persisted bytes; compare naive interpretation with a planned pure subgraph, and wet replay from origin versus checkpoints. Compare coarse preview with declared export resolution. A missed latency, seam, conservation, or reproducibility target falsifies **that route/settings**, not the broader all-brush-graph idea.